

Parallel-element liner impedances for improved absorption of broadband sound in ducts

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(Received 1995 March 21; revised 1995 October 13)

Zwikker and Kosten's theory for propagation of sound in channels was used to predict surface impedance and the spectra of sound absorption for parallel-element acoustical liners. Liner configurations ranged from lumped-element structures with thin acoustically resistive face sheets to ceramic honeycomb structures for which the sub-millimetre-diameter channels provided distributed resistance. Excellent agreement was observed between measured and predicted normal-incidence impedances for constant-depth and variable-depth ceramic-honeycomb structures. For single-degree-of-freedom, lumped-element absorbers, the experimentally observed increase in resistance at frequencies near anti-resonances was predicted. It has been unnecessary to include viscous dissipation near the cavity walls to adequately model the impedance well away from anti-resonance. Inclusion of cavity-wall dissipation allows prediction of the increased resistance at frequencies near anti-resonance. Prediction of this behavior in the region of anti-resonance frequencies may be a critical consideration if parallel elements are to be exploited for increased attenuation bandwidth. © 1995 Institute of Noise Control Engineering.

Primary subject classification: 37.1; Secondary subject classification: 26.1.4

1. INTRODUCTION

The need to reduce noise emission from aircraft operating near airport communities continues to motivate intensive research to improve the sound suppressing efficiency of acoustically treated turbomachinery ducts carrying high-speed subsonic airflows and multiple high-order acoustic modes. Excellent reviews of this effort up to the mid 1970s are given in the literature.¹⁻⁴ The efforts in the 1970s and 1980s resulted in significant progress toward understanding some of the more-important features of sound attenuation in acoustically treated aircraft-engine ducts. Greater attenuation efficiencies are needed to meet future noise reduction goals. Success in attaining these goals will depend, in part, on clever manipulation of the duct wall impedances. New liner configurations that may allow implementation of desired surface impedance distributions are the subject of this paper.

Aircraft duct-liner designs for turbofan engines are required to be concerned with safety, durability, and aerodynamic performance, along with the need to attenuate discrete-frequency sound caused by interactions between the wakes of fan blades and outlet guide vanes. This sound, with spectral peaks at the fan-blade passage frequency and its harmonics, typically exhibits a narrow-band character.

Discrete-frequency fan sound may be suppressed by sound-absorbing liners consisting of thin porous sheets bonded to honeycomb cores^{1,2} as illustrated in the sketch of a test specimen shown in Fig. 1. Such liners are characterized as lumped-element, single-degree-of-freedom (SDOF) resonating systems. The porous sheets provide the dominant portion of the acoustical resistance; the honeycomb

core provides most of the acoustical reactance. Because the individual honeycomb cells are usually acoustically isolated from one another, these liners exhibit locally reacting behavior. The ability to "tune" the attenuation enables the designer to make the liners effective for attenuation of relatively narrow-band noise. The resulting attenuation may not, however, be sufficient for broadband fan noise from future high-bypass-ratio engines or for the discrete-

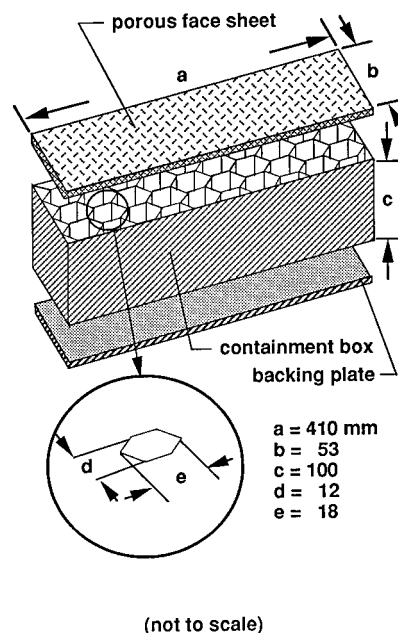


Fig. 1 – Conventional single-degree-of-freedom (CSDOF) liner test specimen construction. The fibermetal porous face sheet was adhesively bonded to the honeycomb partitioned backing cavity.

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frequency fan sound over a wide range of engine power settings.

Broadband noise attenuation in ducts can be achieved by nonresonant liners. For air conditioning ducts, this objective is typically achieved by bulk absorbers. For aircraft applications, however, safety, durability, and structural concerns have precluded use of bulk materials. A different approach is to configure lumped-element resonant systems to achieve broadband attenuation. Exploration of this possibility requires accurate impedance models to handle the more-complicated liner configurations and was the motivation for the work presented in this paper.

This paper documents progress toward an improved understanding of locally reacting acoustical liners that incorporate special features⁵ to manipulate surface impedance distribution and, thus, sound-absorption bandwidth. Previous acoustical impedance measurements of ceramic honeycomb, a simple parallel-element structure consisting of sub-millimetre-diameter cylindrical channels embedded in a solid ceramic matrix (see Fig. 2), indicated that such structures may offer a means to control frequency-dependent impedance behavior.⁶

The predictable internal damping provided by the sub-millimetre channel structure can circumvent the use of resistive face sheets with their attendant mechanical complications, acoustical nonlinearity, and grazing flow effects.⁷ However, ceramic honeycomb, in its current state of development, is too heavy and fragile to be of practical use for aircraft duct-liner applications. These structures can, however, serve as useful tools for liner-concept testing and development, particularly for elevated temperature environments such as the interior of a tailpipe between the turbine

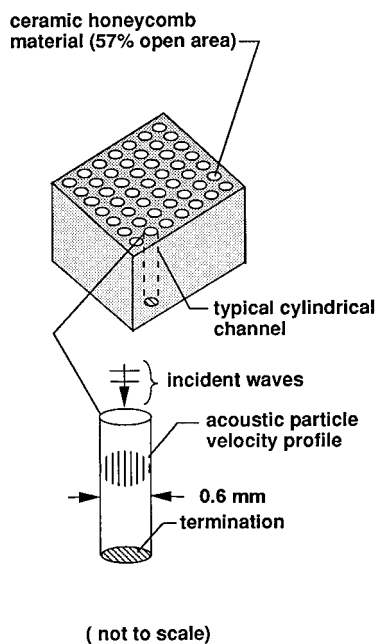


Fig. 2 – Illustration of ceramic honeycomb material with acoustic resistance resulting from energy dissipation by viscous effects within the cylindrical channels.

exhaust and the nozzle exit or the interior of external ejectors in combination with an external mixer nozzle.

This study resulted in the development of a software code that is capable of computing normal-incidence surface impedances for a variety of locally reacting liner configurations. In particular, the code will calculate the individual input impedances of an arbitrary set of channels. Each channel may consist of an arbitrary distribution of lumped and distributed elements. The code sums the area-weighted acoustic admittances of all channels to achieve an effective admittance representative of the admittance of the entire area distribution of channels.

The code was written in a matrix format whereby individual elements making up a single channel can be conveniently combined in series. An important feature of the code is the modularity associated with the matrix formulation. This modularity allows easy revision of channel-element models. The code was designed to efficiently perform parametric and optimization studies of liner configurations to achieve target impedances.

2. SYMBOLS

b	channel depth (m)
b_i	i th segment of cavity depth (m)
c	sound speed (m/s); or channel index
C_p	specific heat at constant pressure [J/(kg·K)]
C_v	specific heat at constant volume [J/(kg·K)]
d_1	liner spatial periodicity in Z-direction (m)
d_2	liner spatial periodicity in Y-direction (m)
d_c	channel diameter (m)
D	duct diameter (m)
f	frequency (Hz)
j	$\sqrt{-1}$
$\mathbf{k}$	propagation vector (m^{-1})
$\underline{k}$	free-space wave number, ω/c (m^{-1})
k	reduced frequency $kd_c/2$
k_z	Z component of wave number (m^{-1})
M	mean flow Mach number
n	number of distinct elements comprising a channel
N_{span}	number of channels in spanwise direction
NSP	number of spans
$p(x_n)$	acoustic pressure at location x_n , normalized by ρc^2
R_f	flow resistance [mks rayls or Pa/(m/s)]
R_1	flow resistivity (mks rayls/m)
s	shear wave number; or span index
$T_{11}, T_{12}, T_{21}, T_{22}$	transmission matrix elements
$u(x_n)$	acoustic particle velocity at location x_n , normalized by sound speed c
X, Y, Z	coordinate directions along liner depth, span, and length (m)
α	absorption coefficient; or real part of propagation constant
β	global surface admittance $1/\zeta$, normalized

	by ρc ; or imaginary part of propagation constant
β_s	parallel admittance for single span of channels, normalized by ρc
β_s^c	admittance for c th channel in s th span, normalized by ρc
χ	acoustic reactance, normalized by ρc
χ_{cav}	channel (or cavity) acoustic reactance, normalized by ρc
χ_{fs}	face-sheet acoustic reactance, normalized by ρc
ϕ	acoustic ray incidence angle (deg)
Γ	propagation constant, $(\beta - j\alpha)$
γ	specific heat ratio, C_p/C_v
κ	thermal conductivity of gas in channel [W/(m·K)]
μ	coefficient of viscosity of gas in channel (N·s/m ²)
θ	acoustic resistance, normalized by ρc
θ_{fs}	face-sheet acoustic resistance, normalized by ρc
ρ	gas density (kg/m ³)
ρ_s	static gas density (kg/m ³)
σ	square root of Prandtl number
Ω	overall open area ratio or porosity of liner segment
Ω_s^c	open area ratio or fraction of the total liner segment porosity attributed to the c th channel
λ_Z	trace wavelength along liner surface in the Z -direction, $\lambda_Z = 2\pi/k_Z$ (m)
ω	angular frequency, rad/s
ζ	liner normal-incidence acoustic impedance, normalized by ρc
ζ_c	characteristic impedance, normalized by ρc
ζ_{cav}	channel (or cavity) acoustic impedance, normalized by ρc
ζ_{fs}	face-sheet acoustic impedance, normalized by ρc
ζ_{opt}	optimum acoustic impedance, normalized by ρc
ζ_s^c	surface impedance for an individual channel, normalized by ρc

Unless otherwise noted, all impedances assume the $e^{+j\omega t}$ time convention for sinusoidal signals.

3. LINER IMPEDANCE PREDICTION

Practical considerations such as structural integrity, weight minimization, durability, and material and fabrication costs have restricted aircraft turbomachinery duct liners mainly to perforated metal sheets or screen-on-perforate face sheets adhesively bonded, brazed, or welded to honeycomb cores. These single-degree-of-freedom (SDOF) absorbing structures can be designed to preferentially suppress discrete frequency sound—the most-significant contributor to the aircraft turbomachinery noise at landing-approach engine power settings.⁸

Current liner design models capture the essential physics of these systems as a lumped resistive element, supplied by the thin face sheet, in series with a reactance supplied by a partitioned backing cavity (see Fig. 1). Generally, dissipation of acoustic energy in the partitioned cavities [usually partitioned with honeycomb structures with cell sizes of at least 6.4 mm (0.25 in.)] is assumed to be negligible relative to the energy dissipated in the resistive face sheet. This assumption is based on the fact that cavity dissipation occurs mainly in the viscous layer at the cell walls. The thickness of this viscous layer is about 0.3 mm for a grazing-incidence 1-kHz sound wave on a rigid wall. Thus, the backing cavities supply essentially a pure reactance and the total normal-incidence impedance ζ is obtained by adding the face sheet impedance to that of a honeycomb cavity as

$$\zeta = \zeta_{fs} + \zeta_{cav} = \theta_{fs} + j(\chi_{fs} + \chi_{cav}), \quad (1)$$

where θ_{fs} is the face-sheet resistance, χ_{fs} is the face-sheet reactance, and χ_{cav} is the cavity channel reactance, all normalized by the acoustic impedance ρc of air at the prevailing atmospheric pressure and air temperature.

The assumption of no dissipation of acoustic energy in the channel of a backing cavity means that the channel impedance is obtained from plane-wave solutions of the non-dissipative wave equation. With the $e^{+j\omega t}$ time convention for sinusoidal signals, the result is

$$\chi_{cav} = -\cot(kb), \quad (2)$$

where k is the free-space wave number and b is channel depth. Resonance occurs at a frequency such that the combined reactance of the face sheet and backing-cavity channel is zero.

The absorption coefficient α for an acoustic ray incident at angle ϕ on a locally reacting surface with impedance $\zeta = \theta + j\chi$ and underneath a grazing flow of Mach number M is given⁹ by

$$\alpha = \frac{4\theta(\cos \phi)(1 + M \sin \phi)}{[1 + \theta(\cos \phi)(1 + M \sin \phi)]^2 + [\chi(\cos \phi)(1 + M \sin \phi)]^2}. \quad (3)$$

Equation (3) should hold approximately for duct geometries such that a sound ray makes a single encounter with the absorbing surface at an angle of incidence ϕ well away from the 90-deg grazing incidence. It is evident that, at a given angle of incidence and with the optimum resistance θ

for maximum sound absorption, broadband absorption is enhanced by keeping the reactance χ as near zero as possible over the frequency range of interest.

From a mode propagation viewpoint, Cremer (as quoted in Ref. 8) showed that the optimum impedance for the

least-attenuated mode in an infinitely long cylindrical duct is given by

$$\zeta_{\text{opt}} = (0.92 - j0.77)kD, \quad (4)$$

where D is the diameter of the duct. Thus, when duct modes are considered, both components of the optimum liner impedance should change linearly with wave number k in contrast to the ray acoustics analysis, which does not account for wave number effects. Although the modal analysis approach is more relevant to the aircraft duct acoustics problem with airflow over the surface of the liner opposite to, and in the direction of, sound propagation, the simpler ray acoustic result was used as a screening tool to select potentially useful liner configurations.

From Eq. (3), ray theory predicts that maximum absorption occurs at zero reactance. For SDOF liners, zero reactance occurs at the resonance frequencies; i.e., when $\chi_{\text{fs}} + \chi_{\text{cav}} = 0$. For resonant liners, the bandwidth for efficient absorption is usually fairly narrow and centered around the resonance frequencies, especially if the face-sheet resistance is chosen to be the optimum. As the resistance deviates from the optimum, some increase in absorption bandwidth can be expected, but at the expense of decreasing absorption in the vicinity of resonances. The face sheet also makes a small positive contribution to the reactance, which is approximately proportional to frequency. This face-sheet reactance effect shifts the liner resonance frequencies to lower values relative to the frequencies of the backing cavity resonances. This effect becomes more pronounced for the higher harmonics of the fundamental liner resonance frequency. Equation (4), for the optimum impedance of the fundamental mode, indicates that the reactance should become more negative with increasing frequency. This impedance behavior is generally unachievable with a SDOF resonant liner configuration.

The above observations suggest that large values of reactance for a SDOF liner near anti-resonance will inhibit the achievement of broadband absorption. To enhance broadband absorption, we seek new, realizable liner configurations that provide a more-desirable variation of the reactive component with frequency. Several liner configurations were explored that, to some extent, exhibited the preferred behavior.

To increase liner absorption bandwidth and efficiency while maintaining viability in realistic operating environments, departures from conventional liner configurations were sought. The design of these liners must also include possible effects of grazing flow and high sound pressures. In addition to the ceramic honeycomb material described earlier, liner configurations for concept development included glass ceramic foams that can be tailored or shaped to fill, completely or partially, aluminum honeycomb channels (with or without face sheets), and structures with variable-depth, buried septa consisting of fibermetals or perforates.

Buried septa divide the backing cavity channels into coupled resonant systems with the septa resistances serving as coupling elements and, thus, provide additional degrees-of-freedom. Septa depth variation from channel-to-channel provides another design parameter that may generate a composite impedance with a greater absorption bandwidth,

i.e., with a less dominant $\cot(kb)$ behavior. The analysis presented in this paper exploits well-known matrix techniques to systematically calculate the composite impedance of liner configurations comprised of contiguous arrays of multi-degree-of-freedom (MDOF) liner elements.

A. Liner impedance calculation formalism

For this investigation, parallel-element liner configurations were treated as acoustically independent (i.e., isolated) channels that, on a local scale, provided incremental variations in the impedance along the liner surface in a prescribed manner. The impedance variation was assumed to be on a spatially periodic scale such that the effective, or "smeared," impedance over the entire liner was a single value obtained from the sum of channel admittances over one spatial period.⁶ Figure 3 depicts such a spatially periodic liner for laboratory testing in a flow-duct arrangement. The test liner, having spanwise and lengthwise spatial periods of d_1 and d_2 , respectively, is pictured as a section of a flow-duct wall. A plane sound wave with a propagation vector $\mathbf{k}$ incident at an angle ϕ in the X-Z plane generates a trace wavelength λ_z corresponding to the trace wave number k_z . Experience¹⁰ suggested that the admittances of the liner channels over one spatial period could be added directly to give an effective admittance, provided that the trace wavelength along the liner was at least eight times that of the liner spatial periodicity; i.e., provided $\lambda_z/d_2 \geq 8$.

The local surface impedance contributed by a particular channel, consisting of n distinct elements in series, is depicted in Fig. 4. The total channel impedance was computed by starting with the boundary condition at the termination of the channel (at x_0 in the figure) and working to the liner surface. The computation was accomplished as a stepwise process utilizing transmission matrices for the various elements comprising the channel. The element

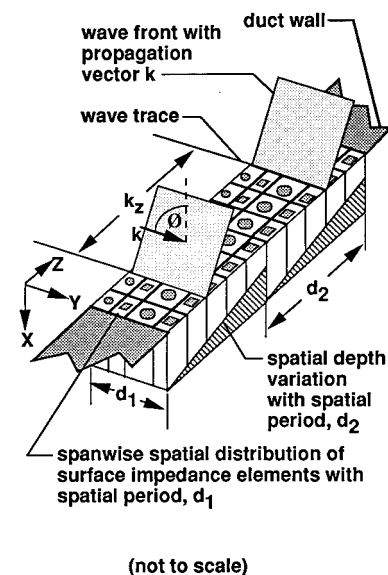


Fig. 3 – Sketch illustrating a locally reacting duct liner with spatial impedance variation in two principal directions.

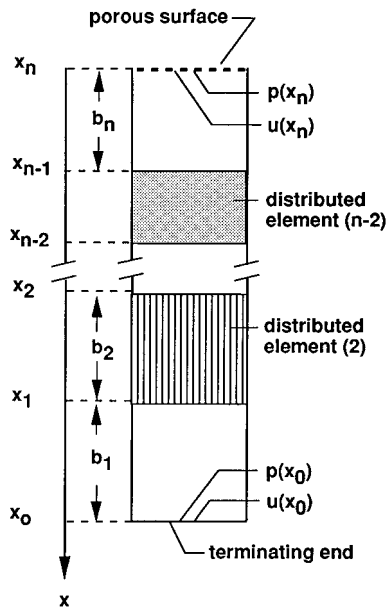


Fig. 4 – Labeling scheme for elements in a typical duct liner with n elements.

transmission matrices were derived analytically, when possible from first principles, or, when necessary, from semi-empirical models.

The terminating condition for an individual liner channel was typically assumed to be a rigid, impervious boundary. It was specified in matrix form as

$$\begin{pmatrix} p(x_0) \\ u(x_0) \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad (5)$$

which yields an infinite impedance. If the channel is terminated in a finite impedance, then appropriate pressures and particle velocities can be assigned to produce the desired terminating impedance.

The forward transmission matrix that describes wave transmission through a dissipative channel takes the following form:¹¹

$$\begin{pmatrix} p(x_{n+1}) \\ u(x_{n+1}) \end{pmatrix} = \begin{pmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{pmatrix} \begin{pmatrix} p(x_n) \\ u(x_n) \end{pmatrix}, \quad (6)$$

where $T_{11} = T_{22} = \cosh(k\Gamma b)$, $T_{12} = \zeta_c \sinh(k\Gamma b)$, and $T_{21} = \zeta_c^{-1} \sinh(k\Gamma b)$. All acoustic pressures and particle velocities are normalized by ρc^2 and c , respectively, where the density of the air and the speed of sound are for the prevailing static pressure and air temperature. Also, impedances and admittances are normalized by the characteristic impedance of air ρc .

Various analytical models are available from the literature to calculate the characteristic impedance ζ_c and the propagation constant Γ . These models are discussed in greater detail below. If the length of an element of the matrix is small relative to the wavelength, the element is said to be a lumped element; i.e., the wave motion through the element can be neglected. The values of the transmission matrix for a purely resistive element can be approxi-

mated by $T_{11} = T_{22} = 1$, $T_{12} = R_f/\rho c$, and $T_{21} = 0$, where the normalized flow resistance $R_f/\rho c$ is usually estimated from a semi-empirical model.

Any number of distributed- or lumped-element transmission matrices can be combined to produce a final input acoustic pressure and velocity. Thus, the impedance ζ_s^c of a single channel with n sub-elements is determined from

$$\zeta_s^c = \frac{p(x_n)}{u(x_n)}. \quad (7)$$

To compute the effective impedance of multiple channels acting in parallel, it was convenient to work directly with admittance $\beta = 1/\zeta$. Within this approximation, the admittance of a liner segment corresponding to spatial periods in the spanwise and lengthwise directions was calculated by combining the channel admittances in parallel.

The effective admittance β of a liner segment is given by

$$\beta = \sum_{s=1}^{NSP} \sum_{c=1}^{N_{span}} \Omega_s^c \beta_s^c, \quad (8)$$

where Ω_s^c is the ratio of “soft to hard surface area” for the c th channel and β_s^c is the admittance associated with the “soft” area of that channel located in the s th span. Equation (8) accounts for any general distribution of channels in backing cavities where the channel admittance and communicating area to the liner surface, associated with each channel, may vary from channel to channel.

For the liner specimens discussed in this paper, area ratio Ω_s^c was held constant throughout the liner configuration and normalized admittance β_s^c was held constant in the spanwise direction, leaving β_s^c to vary only in the lengthwise direction. Thus, Eq. (8) reduced to

$$\beta = \Omega \sum_{s=1}^{NSP} \beta_s, \quad (9)$$

where

$$\beta_s = \sum_{c=1}^{N_{span}} \beta_s^c. \quad (10)$$

In Eq. (9), Ω is the overall open area ratio, or porosity, over a liner spatial period. If the face-sheet porosity is continuously distributed, as might be the case with a fibermetal face sheet, then the elemental porosities Ω_s^c are taken to be 1 (i.e., the liner surface has no impermeable or acoustically hard surface area).

B. Channel sub-element propagation constants

This subsection presents three models for sound transmission in a channel with viscous dissipation taken into account. The models were the basis for transmission matrices useful for distributed elements. The first two models are purely analytical and closely related. They deal with viscous dissipation in a channel filled only with air. The third model is semi-empirical and deals with a porous bulk material filling the channel. For an air-filled channel, the two

different mathematical models arise from different assumptions about the relative volume of the channel affected by viscous dissipation.

Acoustic dissipation in an air-filled channel is confined to a viscous boundary layer induced by the no-slip boundary condition at the wall. The thickness of the frequency-dependent boundary layer is governed by the vorticity diffusion equation.¹¹ For small-diameter channels (typically sub-millimetre and for the frequencies of audible sounds), the “acoustic boundary layer” fills the entire channel. For example, at 1 kHz, the thickness of this viscous layer is about 0.3 mm for a grazing-incidence sound wave on a rigid wall.

As the diameter of the channel increases for a given frequency, viscous dissipation becomes less important and the numerical calculations become cumbersome. Therefore, for “wide” channels, a simplified model is adequate. The viscous boundary layer discussed here is not to be confused with the viscous boundary layer of steady flows, although the physical basis for the two is the same, i.e., viscosity induced no-slip boundary condition at the wall. To distinguish between the two viscous boundary layers, some researchers prefer the term “acoustic boundary layer,” although there is nothing particularly acoustical about it. A more descriptive term might be “viscous penetration depth” as used by Tempkin.¹¹ In this paper, we use the term “viscous boundary layer” as there is, in the present context, no possibility for confusion with the steady-flow boundary layer concept.

The third model describes transmission through a channel filled with a porous bulk material. For this case, boundary layer dissipation at the channel wall (within the bulk material) is completely ignored because such channels, of necessity, will be “wide,” and the dissipation associated with the bulk material will dominate that caused by dissipation in the viscous boundary layer.

In each case, the transmission matrices for the different channel dissipation models described above are determined by the propagation constant and the characteristic impedance. The other transmission matrix of interest to this paper was given in Sec. 3 A for an acoustically thin, resistive element.

The propagation of sound in gases confined in cylindrical tubes has been studied since the latter part of the 18th century. In 1975, Tijdeman¹² concisely summarized the results of those investigations. The most significant solutions for the propagation constant could be expressed in terms of four dimensionless parameters which are, in Tijdeman’s nomenclature: the reduced frequency k , shear wave number s , Prandtl number σ^2 , and specific heat ratio γ , as follows:

$$\bar{k} = \omega d_c / 2c, \quad (11)$$

$$s = \frac{d_c}{2} \sqrt{\frac{\rho_s \omega}{\mu}}, \quad (12)$$

$$\sigma^2 = \mu C_p / \kappa, \quad (13)$$

and

$$\gamma = C_p / C_v. \quad (14)$$

Kirchhoff was apparently the first to obtain a complete analytical solution that included effects of thermal as well as viscous losses. His solution was a complicated transcendental equation, to which he found an approximate solution that holds for “wide” tubes. Rayleigh found an approximate solution to the Kirchhoff equation for “narrow” tubes. Other investigators developed various approximate solutions, either from working with approximate solutions to the Kirchhoff equation or by incorporating more-restrictive assumptions into the fundamental equations.

Tijdeman noted that an analytical solution to the simplified basic equations was obtained by Zwikker and Kosten¹³ in 1949 for the low-reduced-frequency case; i.e., when $k \ll 1$ and $k/s \ll 1$. The solution by Zwikker and Kosten covers the ranges of reduced frequency and shear wave number previously published in the literature. Tijdeman also expressed the Kirchhoff equation in terms of the above four parameters and solved this equation numerically for unrestricted ranges of reduced frequency and shear wave number. He used as a first estimate in his numerical procedure the low reduced-frequency solution of Zwikker and Kosten. Numerical values of the propagation constant are provided in tabular form in Ref. 12 for a wide range of the parameters k and s .

Here, the low-reduced-frequency solution obtained by Zwikker and Kosten was used for the propagation constant, as opposed to the more-exact numerical solution obtained by Tijdeman. This solution appeared adequate for the range of reduced frequency and shear wave number encountered in the proposed liner elements of interest. With Tijdeman’s notation, the Zwikker and Kosten low-reduced-frequency solution for the propagation constant is

$$\Gamma = \sqrt{\frac{J_0(j^{3/2}s)}{J_2(j^{3/2}s)}} \sqrt{\frac{\gamma}{n}}, \quad (15)$$

where

$$n = \left[1 + \frac{\gamma - 1}{\gamma} \frac{J_2(j^{3/2}\sigma s)}{J_0(j^{3/2}\sigma s)} \right]^{-1} \quad (16)$$

and the characteristic impedance is

$$\zeta_c = \frac{-j J_0(j^{3/2}s)}{\Gamma J_2(j^{3/2}s)}. \quad (17)$$

As the channel diameter becomes relatively large compared with the thickness of the viscous boundary layer (or viscous diffusion wavelength), calculation of the required Bessel functions with large arguments becomes difficult. Therefore, in this work, the Kirchhoff “wide-tube” solution was used for shear wave numbers greater than 120; e.g., the shear wave number is 123 for a 12-mm diam tube, driven at 1 kHz. The “wide-tube” solution, from Tijdeman, is

$$\Gamma = \left\{ \frac{1}{\sqrt{2}} \left(\frac{\gamma - 1 + \sigma}{\sigma s} \right) + j \left[1 + \frac{1}{\sqrt{2}} \left(\frac{\gamma - 1 + \sigma}{\sigma s} \right) \right] \right\} \quad (18)$$

and the characteristic impedance is

$$\zeta_c = j/\Gamma. \quad (19)$$

If a channel is filled with a porous material to a depth sufficient that the channel qualifies as an extended propa-

gation medium (i.e., a significant portion of a wavelength), then a propagation constant must be estimated for that region. In the case of fibrous, semi-rigid materials, the Delaney–Bazley semi-empirical formulas¹⁴ for the propagation constant and characteristic impedance are often employed. These formulas are

$$\Gamma = \left[0.189 \left(\frac{\rho f}{R_1} \right)^{-0.593} \right] + j \left[1 + 0.0978 \left(\frac{\rho f}{R_1} \right)^{-0.7} \right] \quad (20)$$

and

$$\zeta_c = \left[1 + 0.0571 \left(\frac{\rho f}{R_1} \right)^{-0.754} \right] - j \left[0.087 \left(\frac{\rho f}{R_1} \right)^{-0.732} \right], \quad (21)$$

where R_1 is the flow resistivity of the material.

The frequency parameter $\rho f/R_1$ in Eqs. (20) and (21) is restricted to the range from 0.01 to 1.00. For some materials, the manufacturer will supply flow resistance data at 20 °C. If the liner is to be designed for high-temperature applications, then all relevant parameters affected by temperature must be appropriately adjusted. Equations (20) and (21) were the basis for the computer code that allowed computation of the “effective” surface impedance of the parallel-element structures discussed in this paper.

4. EXPERIMENTAL SETUP

The impedance data presented in this paper were measured in a normal-incidence impedance tube in the NASA Langley Flow Impedance Test Laboratory. Impedances were measured by the multi-point method.¹⁵ Figure 5 shows a diagram of the normal-incidence tube along with the instruments used. The tube is approximately 0.6-m long with a 51-mm by 51-mm square cross section. Two 120-watt, phase-matched acoustical drivers generated plane sound waves in the tube over a frequency range from 0.3 to

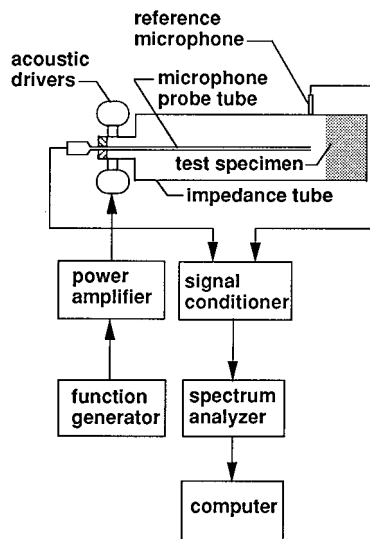


Fig. 5 – Diagram of instrumentation for measurements of acoustic impedance by the “multi-point” method.

3.0 kHz, with sound pressure levels up to 140 dB at the surface of a test specimen. Test specimens were installed in an enclosure that was fitted with a thin rubber gasket, to achieve an airtight seal, then aligned and clamped to the tube exit.

Acoustic plane waves propagate down the tube to the face of the test specimen. Reflected waves create a standing wave pattern that uniquely characterizes the normal-incidence acoustic impedance of the specimen. A 6-mm-diam capacitor microphone, flush mounted in the top wall and 6.4 mm from the face of the test specimen, was used as the reference microphone to measure the sound pressure level near the face of the specimen. This arrangement allowed the sound-pressure levels at the specimen face to be controlled as the frequency was varied.

An axial traversing microphone probe was located on the centerline of the tube. The axial microphone was moved by means of a computer-controlled stepping motor over an axial distance of approximately 0.3 m, to measure the sound-pressure field at discrete points along the standing wave. Construction of the probe is described in detail in Ref. 15.

Figure 5 also shows a block diagram of the signal-conditioning instruments and the data processing system for the acoustical signals. At each test frequency, a sound-pressure level was set at the reference microphone. Amplitude and phase of the sound-pressure signals were acquired and stored digitally for a pre-selected number of axial locations. The data were then processed to obtain normal-incidence impedances as well as other parameters of interest such as complex reflection factors. The data processing procedure is also described in Ref. 15.

5. TEST SPECIMEN DESCRIPTION

Since 1980, the NASA Flow Impedance Test Laboratory has been utilized to measure impedances of numerous conventional liner test specimens and ceramic honeycomb materials. This paper presents measured and calculated acoustic impedance and absorption coefficients for both types of duct-liner test specimens.

A. Fibermetal face sheet with honeycomb

Impedances were measured for a conventional single-degree-of-freedom (CSDOF) test liner. The test specimen was fabricated as an aluminum honeycomb structure adhesively bonded to a fibermetal face sheet. The face-sheet/honeycomb structure was fitted into a rigid container, as illustrated in the sketch of Fig. 1. The test specimen was approximately 410-mm long, 53-mm wide, and 100-mm deep. The nominal flow resistance of the 0.6-mm-thick fibermetal face sheet was given by the manufacturer as 100 mks rays with a nonlinearity factor of 2.1 (i.e., the ratio of flow resistance for a through airflow speed of 2 m/s to the nominal flow resistance at 0.2 m/s).

An expanded view of the hexagonal honeycomb cell geometry is also shown in Fig. 1. The major dimensions of the nominally hexagonal cells are indicated. The cell cross-section area, wetted perimeter, and equivalent diameter

were estimated to be 136 mm², 45 mm, and 13.2 mm, respectively.

B. Ceramic honeycomb

Two test-specimen configurations were fabricated from ceramic honeycomb material as depicted in Fig. 2. The parallel cylindrical channels were uniformly distributed over a 51-mm by 51-mm square area. The open area ratio was about 57%. The mean channel diameter was approximately 0.6 mm.

One test specimen (CDCH-1) had a constant depth of 83 mm. The second test specimen (VDCH-1) was formed by removing material from the CDCH-1 specimen to form a linear variation of channel depth from 33 to 82 mm. The linear depth variation meant that a subset of the total number of channels had their first anti-resonance frequency in the range from 1.9 to 4.7 kHz, as opposed to the constant-depth configuration, where the first anti-resonance occurred at a frequency near 1.9 kHz. For both test configurations, the channel terminations were sealed with a layer of 0.1-mm-thick aluminum tape. The tape seal was assumed to provide a rigid reflecting surface for each channel and isolation from neighboring channels.

Although currently available ceramic honeycomb is brittle and too dense (specific gravity of about 0.5) for application in aircraft engines, it is, because of its acoustical properties, a good baseline material for validation of a computer code for predicting duct-liner impedance, for liner concept development, and for modeling. The acoustic impedance of ceramic honeycomb has been demonstrated to be predictable, linear, and insensitive to flow.⁷

6. RESULTS AND DISCUSSION

A. Theory and experiments

1. Fibermetal face sheet with honeycomb

Figure 6 shows the normalized resistive and reactive components of the measured and predicted normal-incidence acoustic impedance spectra over the frequency range from 0.5 to 3.0 kHz for the conventional single-degree-of-freedom (CSDOF) test specimen depicted by the sketch in Fig. 1. The impedance prediction, based on the Kirchhoff “wide-tube” solution given in Eqs. (18) and (19), took into account dissipation of acoustic energy near the walls of the honeycomb channels. The cavity depth of 100 mm was chosen such that the frequency range included an anti-resonance at 1.7 kHz.

The noteworthy feature of Fig. 6 is the fact that the resistance was predicted to increase dramatically across the anti-resonance region centered at 1.7 kHz. This trend was confirmed by the measured resistance. At frequencies away from anti-resonance, the measured and predicted resistance spectra were approximated very well by the nominal face-sheet flow resistance of about 0.24 ρc . Likewise, the predicted and measured reactance spectra were in good agreement across the entire frequency range, including the anti-resonance region.

Clearly, the face-sheet flow resistance became a negligible component of the total “effective” resistance at fre-

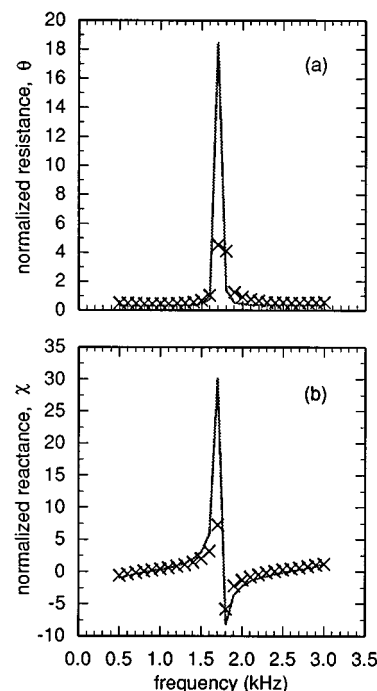


Fig. 6 – Measured (×) and predicted (solid lines) resistive (a) and reactive (b) parts of the normal-incidence acoustic impedance for the conventional single-degree-of-freedom (CSDOF) liner test specimen.

quencies near anti-resonance. This effect was missing completely from the conventional model of thin resistive elements in series combination with partitioned cavities; see Eqs. (1) and (2). Because constant-depth SDOF liner designs with “wide channels” (typically 10-mm cell size) have been the mainstay of duct treatments in the aircraft industry, cavity dissipation has played no role in the total liner impedance. In fact, the “hardening” or increased acoustic resistance of any constant-depth SDOF liner in the vicinity of anti-resonance implies diminished absorption and is of little value to the liner designer. Even though cavity dissipation was included in the predicted behavior, the behavior of the reactive impedance spectrum was still very nearly that of the $-\cot(kb)$ relation as expected for a dissipationless cavity. The measured incremental positive reactance [relative to the $-\cot(kb)$ behavior] with increasing frequency was contributed by the face sheet and was not included in the prediction model.

The observed agreement between measured and predicted impedances in the anti-resonance frequency region for a SDOF liner is encouraging for improved liner designs that may incorporate spatially varying impedance to achieve broader absorption bandwidth. If resonant elements are to be used in this endeavor, then the anti-resonant behavior must be subdued. The following sections describe parallel-element liner configurations that included anti-resonances of individual channels in the frequency range of interest.

2. Constant-depth ceramic honeycomb

Measured and predicted impedance spectra are shown in Fig. 7 for an 83-mm-deep constant-depth ceramic honeycomb structure denoted CDCH-1. This test specimen had no face sheet, and the 0.6-mm channel diameters were about 4% of the nominal size of the hexagonal aluminum honeycomb for which results were shown in Fig. 6. Therefore the resistive component of the surface impedance was dominated by viscous dissipation inside the narrow channels.

Two predictions are compared with the measured impedance. The solid curve (prediction 1) was computed from a propagation constant developed by Kirchhoff, Eqs. (18) and (19), for so-called "wide tubes" and described in Ref. 7. The dashed curve (prediction 2) was computed from a propagation constant given by Zwikker and Kosten; see Eqs. (15) and (16). Although the Kirchhoff based prediction generally did an adequate job, the Zwikker and Kosten result provided a definite improvement, especially at frequencies in the vicinity of resonance. For the predicted results presented in this paper, the propagation constant given by Zwikker and Kosten was used except for those cases where the shear wave number s was so large that the Bessel function arguments became unwieldy to calculate, in which case the Kirchhoff "wide-tube" propagation constant was used.

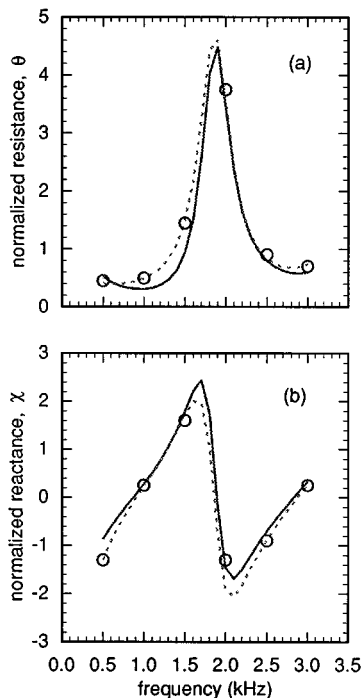


Fig. 7 – Measured (○) and predicted (solid and dashed lines) resistive (a) and reactive (b) parts of the normal-incidence acoustic impedance for the constant-depth ceramic-honeycomb (CDCH-1) liner test specimen. Solid line predictions used the propagation constant developed by Kirchhoff. Dashed-line predictions used the propagation constant developed by Zwikker and Kosten.

It is of interest to compare the impedance spectra of Fig. 6 for the CSDOF liner with the impedance spectra of Fig. 7 for the CDCH-1 liner. The resistance spectrum for the CDCH-1 liner showed a large peak centered at the anti-resonance frequency of about 1.9 kHz. The width of this anti-resonance region ranged from about 1.3 to 2.6 kHz, or about one octave. Also, maximum resistance in the anti-resonance region was less than that for the CSDOF liner. The reactive component behavior departed greatly from the $-\cot(kb)$ relation, although a resemblance was still evident. This result suggested that high values of resistance, associated with anti-resonance for a CSDOF configuration, could be subdued by adding internal dissipation to the liner backing cavity. In the limit of increasing cavity damping and decreasing face-sheet resistance, the impedance spectrum approaches that of a bulk material in contact with a rigid backing with no face sheet; i.e., $\coth(\Gamma b)$.

In what follows, enhancement of broadband absorption for resonant liner structures was attained by increasingly complicated spatial arrays of resonant elements, starting with a variable-depth ceramic-honeycomb structure. Some of the resonant elements of these structures likely were operating near their anti-resonance frequencies. To predict the behavior of these structures, the Zwikker and Kosten channel propagation model was used.

3. Variable-depth ceramic honeycomb

Figure 8 shows the measured and predicted impedance spectra for the variable-depth ceramic-honeycomb test specimen denoted VDCH-1. The measured and predicted

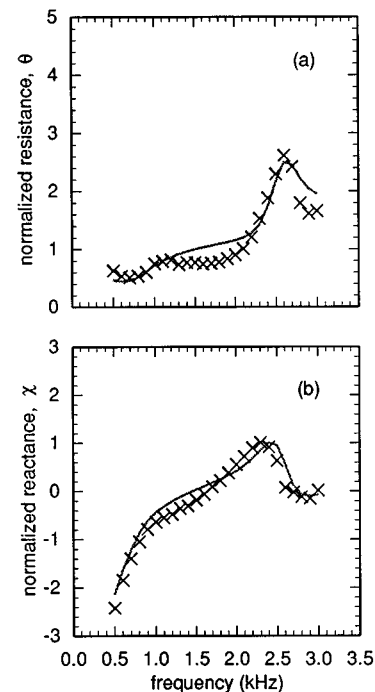


Fig. 8 – Measured (×) and predicted (solid lines) resistive (a) and reactive (b) parts of the normal-incidence acoustic impedance for the variable-depth ceramic-honeycomb (VDCH-1) liner test specimen.

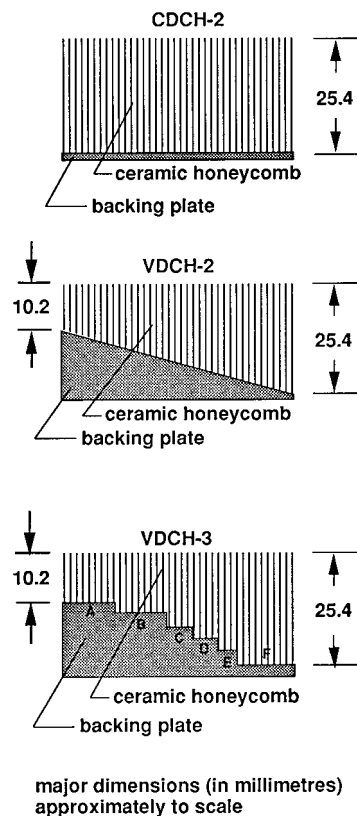


Fig. 9 – Depth profiles in cross sections of ceramic-honeycomb liner test specimens. Constant-depth ceramic honeycomb, CDCH-2. Variable-depth ceramic honeycomb VDCH-2 with linear variation of backing cavity depth. Variable-depth ceramic honeycomb VDCH-3 with stepped distribution of backing cavity depth.

spectra generally were in good agreement. Over the frequency range from about 1.9 to 3.0 kHz, a subset of channels was always operating at anti-resonance. Even so, the Zwikker and Kosten theory predicted the impedance trend in this frequency range. The greatest difference between measured and predicted resistance occurred over the frequency range from 1.2 to 2.2 kHz, where theory overpredicted by at most $0.4 \rho c$. Agreement between measured and predicted reactance was somewhat better than was the case for resistance.

The Zwikker and Kosten theory appeared to be in better agreement with experiment for the CDCH-1 configuration of Fig. 7 than was the case for the VDCH-1 configuration. This slight degradation in predictive ability for the VDCH-1 specimen may be attributed to the presence of interchannel pressure gradients that would tend to exacerbate coupling between channels caused by (1) inadequate channel-to-channel seal provided by the aluminum tape and (2) leakage through the slightly porous channel walls.

The key finding here was experimental validation of the predicted impedance spectrum of a variable-depth honeycomb structure over a frequency range such that some subset of channels may be operating at anti-resonances. Further, by changing the geometry of a test specimen from

uniform depth to a linear depth variation, the resistance and reactance behavior was altered dramatically across the anti-resonance frequency region. The importance of this result is the possibility of manipulating and combining essentially resonant elements in such manner as to extend the absorption efficiency and bandwidth.

B. Further analytical results

Based upon the encouraging results discussed above, the computer code was used to analyze several other interesting liner configuration concepts. Optimization and parametric studies were conducted on these concepts. The results are discussed in the remainder of the paper in terms of absorption coefficient spectra.

1. Ceramic honeycomb depth variation effects

Figure 9 shows sketches of three ceramic honeycomb liner configurations (approximately to scale in overall dimensions) that have the same internal channel structure as for the configuration discussed in Fig. 8. All three configurations had a maximum depth of 25.4 mm and a 51-mm by 51-mm frontal area. They differed only in their terminating geometry. The baseline concept (denoted CDCH-2) had a constant depth of 25.4 mm. Configuration VDCH-2 had a linear depth variation. The third terminating geometry was derived by incorporating the computer code into a routine that optimized on a target maximum absorption and absorption bandwidth. The result was configuration VDCH-3. Six steps were specified in the optimization routine, but variability was permitted in the step depth and width. Table 1 lists the specifications for the optimized stepped-depth distribution of configuration VDCH-3.

Figure 10 shows the predicted absorption coefficient spectra for the three different channel termination geometries. Curve (a), for the CDCH-2 configuration, provides a baseline for comparing the two other depth distributions. Curves (b) and (c) are for the linear (VDCH-2) and optimized depth (VDCH-3) distributions, respectively.

There are several important features to note here. Relative to the CDCH-2 configuration, both the VDCH-2 and VDCH-3 configurations generally provided much greater absorption from about 3.4 to 7.4 kHz. Between 7.4 and 10.0 kHz, the VDCH-2 absorption coefficient reached a minimum of 0.64 at 8.8 kHz. Below about 3.0 kHz, the CDCH-2 configuration had somewhat greater absorption than either the VDCH-2 or VDCH-3 configurations.

TABLE 1 – Optimized distribution of segment widths and depths for the variable-depth, ceramic-honeycomb, parallel-element test configuration VDCH-3. See Fig. 9 for illustration of steps A–F.

Segment	Width (mm)	Depth (mm)
A	10.9	10.2
B	10.9	13.2
C	6.1	16.3
D	6.1	19.3
E	3.3	22.4
F	13.5	25.4

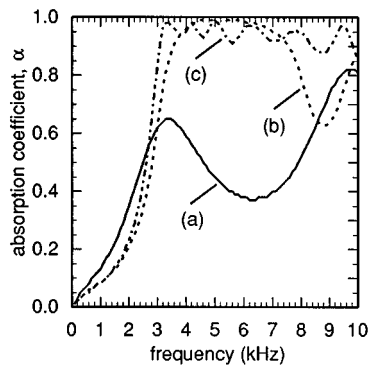


Fig. 10 – Predicted spectra of normal-incidence absorption coefficients for ceramic honeycomb liner test specimens of Fig. 9. (a) Constant depth CDCH-2; (b) linear variable depth VDCH-2; (c) stepped variable depth VDCH-3.

To compare the VDCH-2 and VDCH-3 configurations, the absorption-coefficient bandwidth was defined to be the bandwidth between the limiting frequencies at an absorption coefficient of 0.9. The VDCH-3 absorption bandwidth was 7.0 kHz, whereas the VDCH-2 bandwidth was 4.2 kHz. The increased absorption bandwidth for the VDCH-3 configuration was evidently the result of the relatively small differences (relative to the wavelength at those frequencies) in termination geometry between the two configurations. Careful comparisons of the terminating geometries indicated that the depth distribution for the optimized configuration deviated by at most 4 mm from the corresponding linear depth distribution. The other aspect of the optimized depth distribution was the width assigned to each sub-section. These results suggest that rather significant changes in absorption bandwidth can be attained with minor, but apparently critical, adjustments to a variable-depth-distribution, parallel-element liner.

2. Multi-layer liners

The final part of this section discusses predicted absorption coefficient spectra for four different liner concepts that start with a simple configuration and build up to a multi-layer liner, as illustrated in the sketches of Fig. 11. These sketches were drawn approximately to scale except for the resistive element thicknesses. All configurations were assumed to have a 51-mm by 51-mm frontal area and a total channel length of 76 mm. Channel diameters were set to 7 mm, which is about 11 times the diameters of the ceramic honeycomb channels previously discussed. The wider channel diameter meant that the internal dissipation for these configurations should be much less than that supplied by the smaller ceramic honeycomb channels. Nevertheless, the computer code predicted a significant effect for the internal dissipation, as will be seen.

The configuration geometries depicted in Fig. 11 are: (a) SLO for a single-layer, open-surface liner with no resistive face sheet; (b) SLC for a conventional single-layer liner with a fibermetal face sheet for which the nominal flow

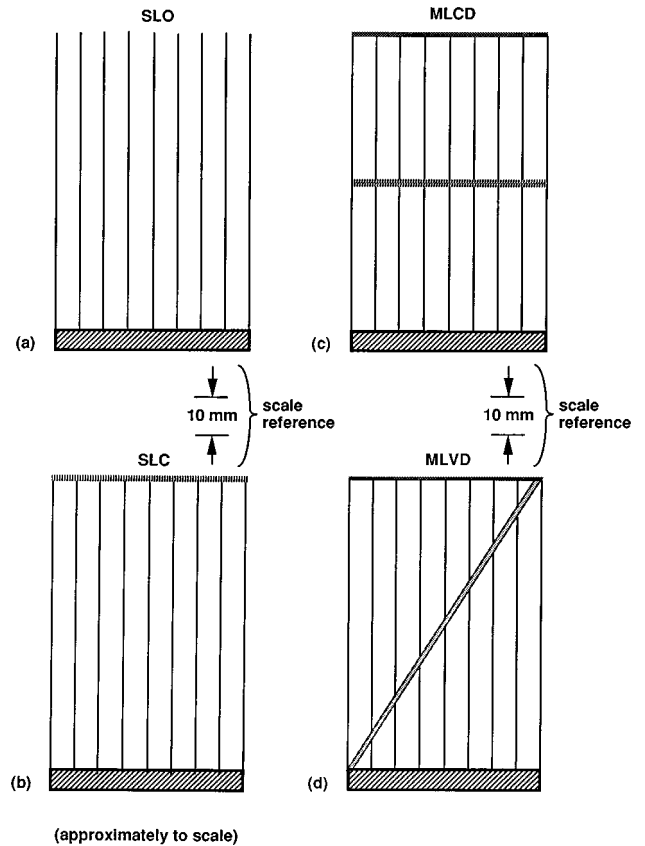


Fig. 11 – Sketches of increasingly complex, locally reacting, resonant-absorber concepts for duct liners. (a) Single layer open (SLO); (b) single-layer closed (SLC); (c) multi-layer, constant depth (MLCD); and (d) multi-layer, variable depth (MLVD).

resistance was 30 mks rayls; (c) MLCD for a two-layer liner formed by adding a thin resistive layer at a constant depth to the SLC configuration of Fig. 11(a); and (d) MLVD for a variable-depth buried-septum liner obtained by setting the resistive layer at a 56-deg angle. The nominal flow resistance of the resistive layers in configurations MLCD and MLVD was 100 mks rayls.

Absorption-coefficient spectra for all four geometries are shown in Fig. 12. The baseline configuration, SLO curve (a), provided a relatively small amount of absorption that increased with frequency to an absorption coefficient of approximately 0.27 at 10 kHz. This absorption was the result of viscous losses near the channel walls, losses neglected in conventional modeling of these structures but included here. From Eq. (12), at 10 kHz, the maximum shear wave number s and reduced frequency k are 228 and 0.64, respectively. These values suggested that the condition $k \ll 1$, assumed for the validity of the Zwikker and Kosten solution, was suspect. However, an examination of tabulated numerical results in Ref. 12 for k values up to 1.57 and s values up to 100 indicated that the Zwikker and Kosten solution remains accurate to the third decimal place.

As expected, the SLC configuration [curve (b) in Fig. 12] provided relatively high absorption in narrow fre-

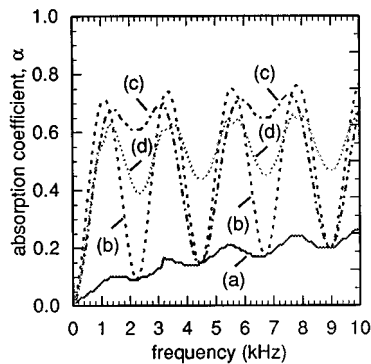


Fig. 12 – Predicted spectra of normal-incidence absorption coefficients for duct-liner concepts illustrated in Fig. 11. (a) Single-layer open (SLO); (b) single-layer closed (SLC); (c) multi-layer, constant depth (MLCD); and (d) multi-layer, variable depth (MLVD).

frequency bands centered on the fundamental resonance frequency of about 1.1 kHz and its odd numbered harmonics. Absorption minima occurring at anti-resonances (starting at 2.2 kHz) were bounded by the open channel absorption of curve (a) for configuration SLO. The absorption minima, corresponding to the anti-resonances for the SLC configuration, tracked the open-channel absorption spectrum. The absorption coefficient maxima increased from about 0.70 to about 0.95 in a monotonic manner.

Of more interest is the behavior of the multi-layer MLCD and MLVD configurations, curves (c) and (d), respectively, in Fig. 12. The MLCD liner “filled in” the low absorption valley associated with the first anti-resonance at 2.2 kHz. At the second anti-resonance near 4.5 kHz, the absorption for the MLCD liner was very nearly identical to that observed for the SLC configuration. This pattern (absorption increases at odd-numbered anti-resonances, matching to SLC absorption at even-numbered anti-resonances) continued for alternating anti-resonances.

For the parameters used in this calculation, the absorption coefficients for the MLCD configuration ranged up to nearly 0.7 in the first high-frequency absorption band from 0.6 to 3.6 kHz, defined at an absorption coefficient of 0.5. In effect, the 3.0-kHz bandwidth of the MLCD liner configuration was three times the corresponding bandwidth for the SLC configuration. This pattern repeated with frequency to define a second absorption band in the frequency range from 4.5 to 9.0 kHz.

The absorption spectrum of the MLVD configuration [curve (d) of Fig. 12] filled in the anti-resonance absorption minima in a more consistent manner than did the MLCD configuration. This result was achieved at the expense of decreased absorption, relative to the absorption achieved by the MLCD configuration, at odd numbered anti-resonances starting at 2.2 kHz. The absorption coefficients for the MLVD configuration varied from about 0.40 to 0.62, with the maximum absorption occurring slightly above the SLC resonances and with the minima occurring near the anti-resonances of the SLC configuration.

Another way to view the relative performances of the

two buried-septa multi-layer liners is that, while both provided improvements over the absorption provided by the SLC configuration, the MLVD configuration apparently was able to provide a much greater absorption bandwidth but with degraded performance relative to the MLCD configuration, especially at what corresponded to the anti-resonance frequencies of the SLC configuration. It should be noted that these results are for one set of resistive layer parameters. New features are certain to arise for other geometric and resistance parameters.

Finally, the MLVD configuration absorption spectra are shown in Fig. 13 for three temperatures 294, 533, and 811 K corresponding to curves (a), (b), and (c), respectively. Note that curve (a) in Fig. 13, for the ambient temperature condition, essentially replicated curve (d) in Fig. 12 to about 6 kHz (or to a reduced frequency of 0.35). In Fig. 13, the absorption coefficient spectra are plotted versus reduced frequency to remove the effect of sound-speed dependence on temperature. The nominal flow resistance of the face sheet and buried septum were the same as for the configurations for which the data are shown in Fig. 12.

For the multi-layer variable-depth liner configuration for which results are shown in Fig. 13, the effect of increased temperature was to decrease the absorption. This outcome may seem counter-intuitive in view of the fact that increasing kinematic viscosity with temperature implies greater acoustic energy attenuation as it propagates through a channel. The results shown in Fig. 13 may be explained by the fact that the surface acoustic impedance relative to the characteristic impedance of the medium changes with temperature. Thus, the absorption of acoustic energy at the liner surface is a function not only of how efficiently energy is dissipated in the interior of the liner but also how efficiently energy is transported across the impedance mismatch at the interface.

7. CONCLUDING REMARKS

A computer code was created that allows prediction of the surface impedance of any series/parallel-element acous-

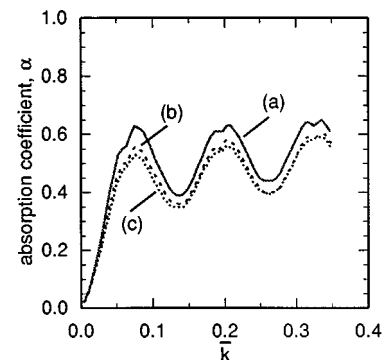


Fig. 13 – Predicted reduced-frequency spectra of normal incidence absorption coefficients for the multi-layer, variable-depth liner (MLVD) at three temperatures. (a) 294 K; (b) 533 K; and (c) 811 K.

tical liner configuration that can be assumed to be locally reacting. Predicted impedances were validated experimentally for selected test specimens. The code was exercised for several other liner configurations to explore possibilities for increasing absorption bandwidth.

Some of the more significant results were:

- Surface impedance predictions, calculated on the basis of the Zwikker and Kosten solution to the Kirchhoff equation for propagation in channels with viscous dissipation, were validated experimentally for absorbing structures that incorporated such channels to partition their backing cavities.

- The experimentally observed increase in resistance at frequencies in the region of anti-resonance for single-degree-of-freedom resonant absorbers was validated experimentally. These absorbers have previously been modeled as lumped-element systems for which no resistance increase in the neighborhood of the anti-resonance frequencies was predicted. The ability to predict the resistance increase may be of critical importance to the exploitation of single-degree-of-freedom parallel elements to achieve broader absorption bandwidth.

- Excellent agreement between experiment and predictions based on the Zwikker and Kosten theory was observed for constant-depth and variable-depth ceramic-honeycomb structures. Normal-incidence absorption coefficients could be increased dramatically by means of spatially distributed parallel elements for which the resonance frequencies differed in a systematic manner by spatially varying the channel depth.

- Parametric and optimization calculations to explore increases in absorption bandwidth for various configurations of channel partitioned cavities, incorporating face sheets and buried septa, yielded encouraging results. The results indicated that a constant-depth buried-septum liner performed better than a variable-depth buried-septum liner over a bandwidth of about 3 kHz centered approximately on the anti-resonances of a single-layer liner of the same total depth. In contrast, a variable-depth buried-septum liner performed better than a constant-depth buried-septum liner, in an average sense, over an extended frequency range. However, at frequencies corresponding to the anti-

resonances of the same-depth single-layer liner, the variable-depth buried-septum liner performance was degraded relative to that of the constant-depth buried septum liner.

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